

Publication-readiness plan for the seeded inverse-designed SiN nanobeam Rb cavity

Datasets, tuning tasks, and figure package for Claude Opus 4.8

Input report: InverseDesign_Comparison_Report.pdf | Date: 2026-06-15

Constraint	Implication for this work
Allowed simulations	Wave-EM eigenmode/time-domain/frequency-domain verification, Bloch band-structure simulations, port-flux calculations, mode-volume postprocessing, geometry sweeps, and simple scalar postprocessing.
Out of scope	Do not implement full cavity-QED, master-equation, quantum-trajectory, atom-motion, Raman/STIRAP, hyperfine-control, or gate-fidelity simulations. Do not require new atomic-physics infrastructure.
Publication target	Support a photonic-design claim: a manufacturable SiN nanobeam cavity geometry with high waveguide extraction at the 87Rb D2 frequency and robust EM-derived interface figures of merit.

1. Current status and immediate publication gaps

The present invDes(seed) design is promising because it combines high waveguide extraction with adequate cooperativity after seeded inverse design. However, the current report is not yet publication-ready because the final design is not exactly resonant with the Rb D2 target and because the supporting datasets do not yet demonstrate tuning, robustness, mode isolation, numerical convergence, or a complete loss budget.

Quantity from current report	invDes(seed)	Publication-readiness interpretation
f _{res}	384.53 THz	Target is 384.23 THz, so the final design is high by 0.30 THz = 300 GHz. With $\kappa/2\pi = 30.5$ GHz, this is about 9.8 loaded linewidths and must be tuned.
eta at atom height 250 nm	0.834	Excellent nominal EM-derived interface efficiency, but it must be recomputed after final resonance trim and over tolerance/position sweeps.
beta = κ_{wg}/κ_{tot}	0.875	The most important success: the port is strongly overcoupled to the desired waveguide. Publication figures should decompose where the remaining 12.5% loss goes.
Q loaded; $\kappa/2\pi$	1.26e4; 30.5 GHz	Bad-cavity extraction regime is plausible. Need independent Q verification, fit residuals, and convergence checks.
C at 250 nm; $g/2\pi$	25.4; 1.08 GHz	Adequate for the EM interface claim, but absolute g conventions need a single documented convention and no overclaiming of qubit fidelity.
Mode volume	$0.86 (\lambda/n)^3$	Competitive if convention, normalization, and spatial field maximum are documented and convergence-tested.

Key diagnosis: The design already looks like a strong EM cavity-to-waveguide interface, but the final publication claim must be built around a resonance-trimmed design family, not only the single 384.53 THz geometry. A simple first-order global scale estimate is: increase all in-plane/3D dimensions by $f_{current}/f_{target} - 1 = 0.078\%$. Because f approximately scales as 1/length, this should move 384.53 THz toward 384.23 THz before local retuning.

2. Tuning needed before declaring a final design

Priority	Tuning task	Implementation using allowed simulations	Success criterion
P0	Final spectral trim to Rb D2	Start from invDes(seed). Run a global geometry-scale sweep around +0.078% and a local defect-only scale sweep. For each candidate, verify f_{res} , Q , κ_{wg} , β , V , C , and η with the same verification pipeline.	Nominal target: $ f_{\text{res}} - f_{\text{D2}} \leq 5 \text{ GHz}$. Hard target: $ f_{\text{res}} - f_{\text{D2}} \leq 0.25 * (\kappa_{\text{tot}}/2\pi)$, whichever is larger.
P0	Rebalance output coupling after trim	Sweep output mirror strength: number of output mirror cells, output-side hole radii, and taper endpoints. Recompute β , Q , κ_{wg} , $\kappa_{\text{rad/back}}$, and η .	$\beta \geq 0.80$ and $\eta \geq 0.75$ after resonance trim; avoid Q collapse below the range needed for $C \geq 20$.
P1	Loss-budget decomposition	Use port fluxes, total stored energy, PML/radiated flux, and back-side leakage estimates. If exact back/radiation split is hard, at least report $\kappa_{\text{wg}}/\kappa_{\text{tot}}$ and residual loss = $1 - \beta$.	A figure must show κ_{wg} , residual loss, and total κ . The extraction claim should be traceable to classical EM power accounting.
P1	Mode isolation and spectral cleanliness	Search for nearby cavity resonances over a broad frequency window around D2 using the same verification method. Plot the chosen mode and nearest competing modes.	Nearest same-polarization/localized resonance should be separated by at least 3 loaded linewidths; preferably $>100 \text{ GHz}$ for simple interpretation.
P1	Geometry smoothness and manufacturability	Flag hole-to-hole jumps in lattice constant, hole radii, and minimum feature sizes. Optionally apply bounded smoothing and reverify.	No isolated high-curvature parameter glitches unless physically justified by the optimizer and robust to fabrication sweeps.
P2	Optional second-stage inverse design	Restart optimization from the best resonance-trimmed candidate with a loss that penalizes detuning, rewards β , and preserves adequate C . Keep the parameterization fabrication-aware.	Only accept if it improves η or robustness without producing unrealistic geometry or fragile spectral behavior.

3. Minimal EM-derived figures of merit

Use one consistent convention across every figure and dataset. Do not mix definitions inherited from earlier reports. The publication can state that the atom-light metrics are EM-derived scalar proxies from standard cavity parameters, not outputs of a full cavity-QED simulation.

Metric	Computation	Notes for Claude
f_{res} and Q	Extract from time-domain ringdown, frequency-domain resonance fit, or eigenmode solver. Use $Q = f_{\text{res}}/(\kappa_{\text{tot}}/2\pi)$.	Save fit window, raw spectra, fitted curve, residuals, and confidence/fit diagnostics when possible.
β	$\beta = \kappa_{\text{wg}}/\kappa_{\text{tot}}$ from waveguide port power divided by total loss rate.	This is the central interface metric. Preserve the sign/port convention and report the output port used.

Metric	Computation	Notes for Claude
Mode volume V	$V = \text{integral} \epsilon E ^2 dV / \max(\epsilon E ^2)$, reported in $(\lambda/n)^3$ using the existing convention.	Also save the field maximum location and atom-sampling point.
g proxy	Use the same field-normalization and dipole convention already used by the repository. Document any factor-of-2 convention explicitly.	Do not introduce a new atomic package. This is a postprocessing scalar from EM fields.
C proxy	Use the existing formula, e.g. $C = 4g^2/(\kappa \gamma)$ if that is the code convention.	Report γ and convention in metadata. Keep comparisons internal and consistent.
eta proxy	Use $\eta = \beta C/(1+C)$ for the resonant one-sided interface proxy, then optionally show Lorentzian detuning suppression as $\eta_{\text{eff}} = \eta/[1 + (2\Delta/\kappa)^2]$.	This is simple scalar postprocessing, not a dynamic atom simulation. Use only as an interpretive correction for detuned candidates.

4. Required datasets for a publication package

Dataset	Files to produce	Minimum contents
Final design geometry package	final_design.json, final_design.gds, final_design.csv	All per-hole parameters, beam width/thickness, coordinate conventions, material indices, target frequency, and output-port location. Include seed and invDes(seed) for comparison.
Nominal verification dataset	nominal_verify.json + fields_final.npz/h5	f_res, Q, kappa_tot/2pi, kappa_wg/2pi, beta, V, g, C, eta, field arrays or sampled field cuts, energy integrals, port fluxes, and simulation settings.
Band-structure dataset	bands_seed.json, bands_final.json, bands_sweep/*.json	Mirror, defect, and output/taper unit-cell bands; bandgap edges; light line; target Rb D2 line; final cavity frequency overlaid.
Resonance-trim sweep	tuning_global_scale.csv/json, tuning_defect_scale.csv/json	Global scale, local defect scale, output-side scale, f_res, Q, beta, V, C, eta, and selected final candidate. Include both raw and interpolated optimum.
Output-coupling sweep	output_coupling_sweep.csv/json	Output mirror cell count/strength, kappa_wg, residual loss, Q, beta, eta. This supports the claim that inverse design opened the correct port.
Atom-position map	position_map_xz.npz, position_map_yz.npz	At minimum, compute $ E $ at atom positions and derive g, C, eta over height 200-350 nm and lateral offsets. Use beta and kappa from the cavity unless re-solving every position is cheap.
Fabrication/tolerance sweep	tolerance_sweep.csv/json + tolerance_samples/	Uniform scale error, hole-radius etch bias, lattice constant error, hole ellipticity, beam width/thickness sweeps if supported. Include simple retuning by global scale if feasible.
Numerical convergence dataset	convergence_mesh.csv/json, convergence_pml.csv/json, convergence_domain.csv/json	Vary mesh resolution, PML thickness, domain size, time window/frequency fit window. Report drift in f, Q, beta, V, and eta.
Mode-isolation dataset	mode_scan.csv/json + competing_modes.npz	Broad frequency scan around D2, nearest competing resonances, Q and field plots for competing modes.
Comparison dataset	comparison_table.csv/json	design_1, invDes(design_1), seed, invDes(seed), final_trimmed_design, and any robustness-adjusted finalist. Same verification pipeline for all.

5. Figures that should appear in the final publication

The main text should be built around a small number of decisive, reproducible figures. Each figure must have a backing dataset in machine-readable form and a script that regenerates it from the saved data.

Figure	What it should show	Why it matters
Fig. 1 - Device and design flow	Top-view final geometry, atom position, output waveguide, mirror/defect/output sections, plus per-hole lattice/radius traces for seed vs final.	Shows that the device is a physically interpretable nanobeam, not an opaque pixel pattern.
Fig. 2 - Band-engineered seed physics	Bloch bands for mirror and defect cells, bandgap shading, light line, Rb D2 target, and final f_{res} overlay.	Demonstrates that the inverse design is seeded by correct photonic-crystal physics.
Fig. 3 - Final resonance and loss budget	Resonance spectrum/ringdown fit, Q , κ_{tot} , κ_{wg} , residual loss, β , and fit residuals.	This is the core proof that the final design is resonant and preferentially emits into the waveguide.
Fig. 4 - Field localization and waveguide extraction	$ E $ or energy-density cuts in the beam midplane and vertical atom plane; waveguide flux direction; field at the atom height.	Shows mode localization, output coupling direction, and field accessibility at the atom location.
Fig. 5 - Tuning and Pareto map	Global-scale/defect-scale/output-coupling sweeps with η , β , Q , C , and detuning. Mark seed, $\text{invDes}(\text{seed})$, and final trimmed design.	Turns the main weakness - detuning - into a demonstrated, quantitative tuning story.
Fig. 6 - Atom-position robustness	Heatmaps of η and C versus atom height and lateral offset, with the nominal 250 nm point marked.	Shows whether the cavity is useful for a trapped atom with finite positioning uncertainty.
Fig. 7 - Fabrication robustness	Histograms or violin plots for η , β , f_{res} detuning, and Q under simple fabrication perturbations; include retuned and unretuned cases if possible.	Publication reviewers will ask whether the design is a fragile numerical artifact.
Fig. 8 - Four-design plus final comparison	Bar/table comparison of design_1 , $\text{invDes}(\text{design}_1)$, seed, $\text{invDes}(\text{seed})$, and final trimmed design for f_{res} , η , β , Q , κ , C , V .	Connects the final design to the existing report and shows what the new tuning contributed.

6. Supplementary figures and tables

- Complete per-hole parameter table for the final design and the seed, including fabrication bounds and units.
- Numerical convergence plots: mesh, PML, domain size, time window, and fitting-window sensitivity.
- Mode-isolation scan with field plots of the nearest competing resonances.
- Full output-coupling sweep and local/global scale sweep, not only the final selected point.
- Raw optimization trajectories with physical metrics per iteration, not just optimizer loss.
- All field-normalization and mode-volume convention details, including a comparison to the legacy convention if necessary.
- GDS rendering and minimum-feature annotation for fabrication review.
- Data and script manifest: exact commands to regenerate every figure from saved data.

7. Specific simple simulations to implement

Simulation/check	Recommended sweep	Implementation notes
Global scale trim	Scale all dimensions by 0.999 to 1.002 in steps of 0.0001 near the predicted +0.00078 point; refine around the crossing.	If thickness cannot be scaled in the current geometry pipeline, do both in-plane-only and full-scale variants and label them clearly.
Defect-only spectral trim	Scale only the central defect/taper region by small factors while holding far mirrors fixed.	This can tune f_{res} with less impact on output β than full scaling; verify both.

Simulation/check	Recommended sweep	Implementation notes
Output mirror strength	Sweep output mirror count/strength around the current 4-cell setting and one or two continuous hole-size/taper parameters.	Look for maximum eta, not maximum Q. Save kappa_wg and residual loss separately.
Back mirror length	Sweep back mirror cell count, e.g. 20, 25, 30, 35, 40 if computationally feasible.	Establish that the back mirror is long enough that extra cells do not meaningfully increase eta.
Atom height/lateral map	z above top surface: 200-350 nm; x and y offsets around nominal field maximum. Use 2D maps and one-line cuts.	This is cheap if based on one solved field and scalar postprocessing.
Etch bias perturbations	Uniform hole radius bias: -5 to +5 nm; lattice bias: -2 to +2 nm; beam width bias if supported.	Do not overcomplicate. A small deterministic grid plus 50-200 random samples is enough for a first publication-quality tolerance picture.
Mesh/PML convergence	At least three mesh resolutions and two PML/domain settings.	A table of relative changes is acceptable if full plots are too large.
Mode scan	Frequency window at least +/- 0.5 THz around Rb D2, broader if cheap.	Confirm no nearby mode invalidates a single-mode cavity interpretation.

8. Acceptance checklist for the final design

- A final geometry exists whose verified resonance is within the defined D2 detuning tolerance.
- The final geometry retains $\beta \geq 0.80$ and $\eta \geq 0.75$ under the same metric convention as the original report, or the report explains the tradeoff quantitatively.
- The loss budget explains the loaded Q and shows the desired waveguide channel dominates the decay.
- The design has a documented band-structure rationale and a documented inverse-design improvement over the seed.
- The design has at least one practical tuning path: global scale, local defect scale, or output-coupling retune.
- The final claims are robust to mesh/PML/domain choices within a small, stated numerical uncertainty.
- The final claims are not destroyed by simple, realistic fabrication perturbations, or the required fabrication precision is explicitly stated.
- All figures are reproducible from saved machine-readable data; no figure depends only on a notebook state or screenshot.
- The manuscript language avoids claiming atom-qubit gate/readout fidelity, entanglement fidelity, or full quantum-network performance because those would require simulations outside the present scope.

9. Suggested Claude Opus 4.8 implementation sequence

Step	Task	Expected output
1	Read existing comparison report outputs and locate final invDes(seed) geometry, verification scripts, band scripts, and record.json files.	Short inventory file: publication_ready/inventory.md.
2	Create a publication_ready module or script folder that can run all publication checks without modifying the core inverse-design code unless necessary.	Scripts: run_tuning_sweeps.py, run_band_figures.py, run_position_maps.py, run_tolerance.py, make_publication_figures.py.
3	Run the global-scale and defect-scale resonance trims first; choose a final candidate based on detuning, eta, beta, and Q.	Dataset: tuning_global_scale.*, tuning_defect_scale.*, final_candidate.json.
4	Run output-coupling sweep around the final candidate if beta/eta degrades after spectral trim.	Dataset: output_coupling_sweep.* and updated final_candidate.json.

Step	Task	Expected output
5	Run nominal verification, field export, loss-budget accounting, and mode-volume/g/C/eta postprocessing for the final candidate.	Dataset: nominal_verify.* and fields_final.*.
6	Generate band-structure figures for seed and final-relevant unit cells.	Dataset: bands_final.* and publication Fig. 2.
7	Run simple robustness studies: position map, tolerance grid/random samples, numerical convergence, and mode isolation.	Datasets and supplementary figures.
8	Generate all publication figures and a final summary table, with a manifest mapping each figure to the exact data file and command.	publication_figures/*.pdf, publication_ready_manifest.md, comparison_table.csv/json.

10. Final output manifest Claude should produce

- publication_ready/final_candidate.json - final selected geometry and metadata.
- publication_ready/data/*.json, *.csv, *.npz - all raw and processed datasets listed above.
- publication_ready/figures/main/Fig1_*.pdf through Fig8_*.pdf - main-text figures.
- publication_ready/figures/supplement/*.pdf - convergence, tolerance, and extended sweeps.
- publication_ready/tables/comparison_table.csv - uniform metrics for all designs.
- publication_ready/manuscript_notes.md - concise narrative of what changed from invDes(seed) to the final trimmed design.
- publication_ready/reproduce_all.sh - one command that regenerates all final figures from saved datasets, without rerunning expensive simulations unless requested.
- publication_ready/limitations.md - explicit statement that the work uses EM-derived interface metrics only and does not include full cavity-QED dynamics.